

IdeaForge AI

Project Blueprint

Playlist Manager

Domain

Music

Problem Statement

Users struggle to manage their large music playlists efficiently.

Proposed Solution

A music playlist manager that allows users to easily organize, search, and share their playlists.

Platform

Web application

AI Suitability

Moderate

LLM Model

llama3.2:3b

Embedding Model

nomic-embed-text

RAG Required

True

RAG Use

For song tagging and filtering

Database

MongoDB

Technical Feasibility

High

Difficulty

Medium

Societal Usefulness

High

Target Users

- Music enthusiasts
- Professional DJs
- Music bloggers

Objectives

- Effortless playlist organization
- Improved music discovery
- Enhanced sharing capabilities

Core Features

- Playlist creation and editing
- Song tagging and filtering
- Recommendation engine
- Sharing and collaboration tools

Recommended Features

- Mood-based playlist creation
- Social features for music sharing
- Personalized recommendations

How It Works

- User input for playlist creation and editing
- Database storage for playlist data
- Algorithmic recommendations for new songs
- Real-time sharing and collaboration

AI Approach

- Natural Language Processing for song tagging and filtering

Technology Stack

- Python
- Streamlit
- FastAPI

- ChromaDB
- MongoDB

Data Requirements

- Song metadata
- User preferences
- Playlist data

Main Modules

- Playlist creation and editing
- Recommendation engine
- Sharing and collaboration tools

Workflow

- User input
- Database storage
- Algorithmic processing
- Real-time sharing

Improvement Ideas

- Integrate social media sharing
- Add personalized music recommendations

Alternative Ideas

- Mobile application
- Voice-controlled playlist manager